

ISLAMIC UNIVERSITY OF TECHNOLOGY(IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

CSE 4603: Software Testing & Quality Assurance
Quiz 3

Time 35 minutes

Marks 15

1. T contains 90 tests of which 20 are modification-revealing for P and P' and M selects 12 of these 20 tests, calculate the inclusiveness of M relative to P, P', and T.
Answer: $m = \text{selected test cases} = 12$; $n = \text{total number of modification revealing test cases} = 20$
 $m/n = 100 * 12/20 = 60\%$
2. How regression testing can pose a problem in a real life situation?
Answer slide 7, page 30
3. Differentiate between Load Testing, Stress Testing and Scalability Testing

Load	Stress	Scalability
To determine the system's stability and robustness by testing it beyond its normal operating conditions.	To measure the system's performance under expected user loads to ensure it can handle high traffic volumes.	To determine the system's ability to scale up or down in terms of user load, data volume, and transaction counts.
Check if the system can handle targeted users	Test to check the failure behavior of the system	Test the system to check the behavior before crashing (corrective measures)
To validate the scalability and reliability of the system.	To check how the system handles failures and recovers from them.	To identify potential scalability issues before they affect the production environment.

4. Imagine a modern e-commerce platform where users land on the home page, view featured products, and can search for specific items. They can add products to their cart, review the cart, proceed to checkout, and receive an order confirmation. Logged-in users have additional features like order history and saved addresses. They can also file complaints for any product. The application is responsive across devices.

In the provided scenario, which modules would you like to test to ensure the application's performance?

Answer:

High Frequency modules: add product, search, view page, home page

Most critical modules: proceed to checkout

Read Transaction: Order history

Update Transaction: review cart